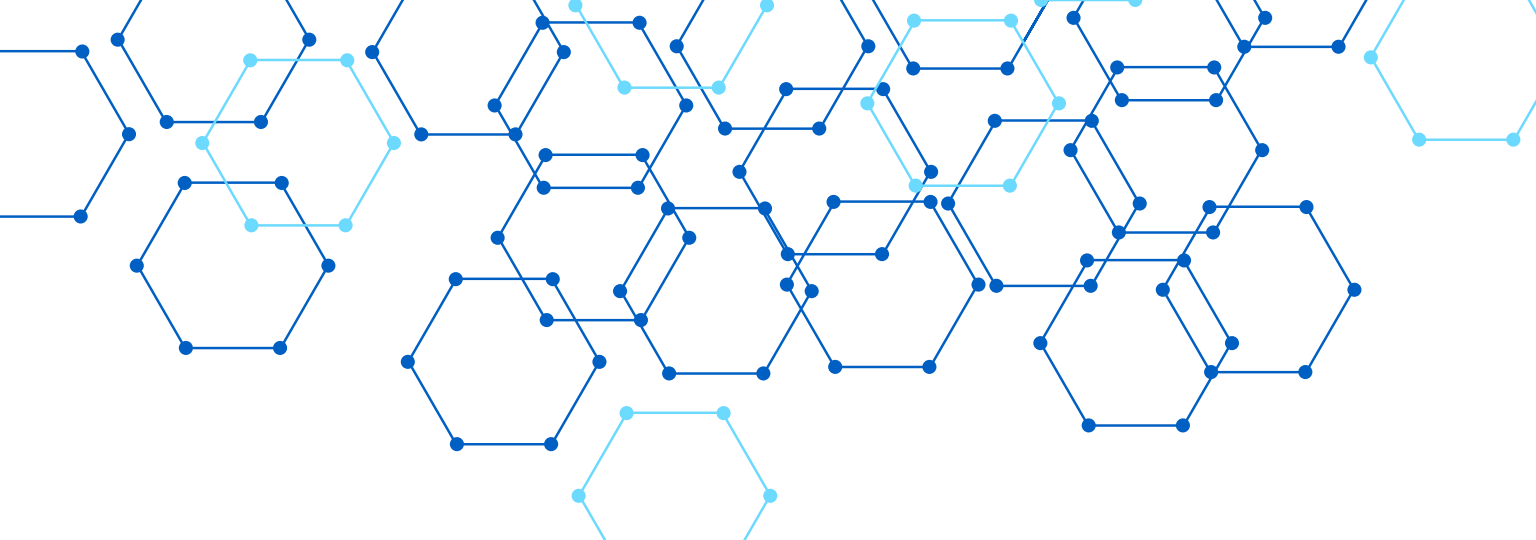
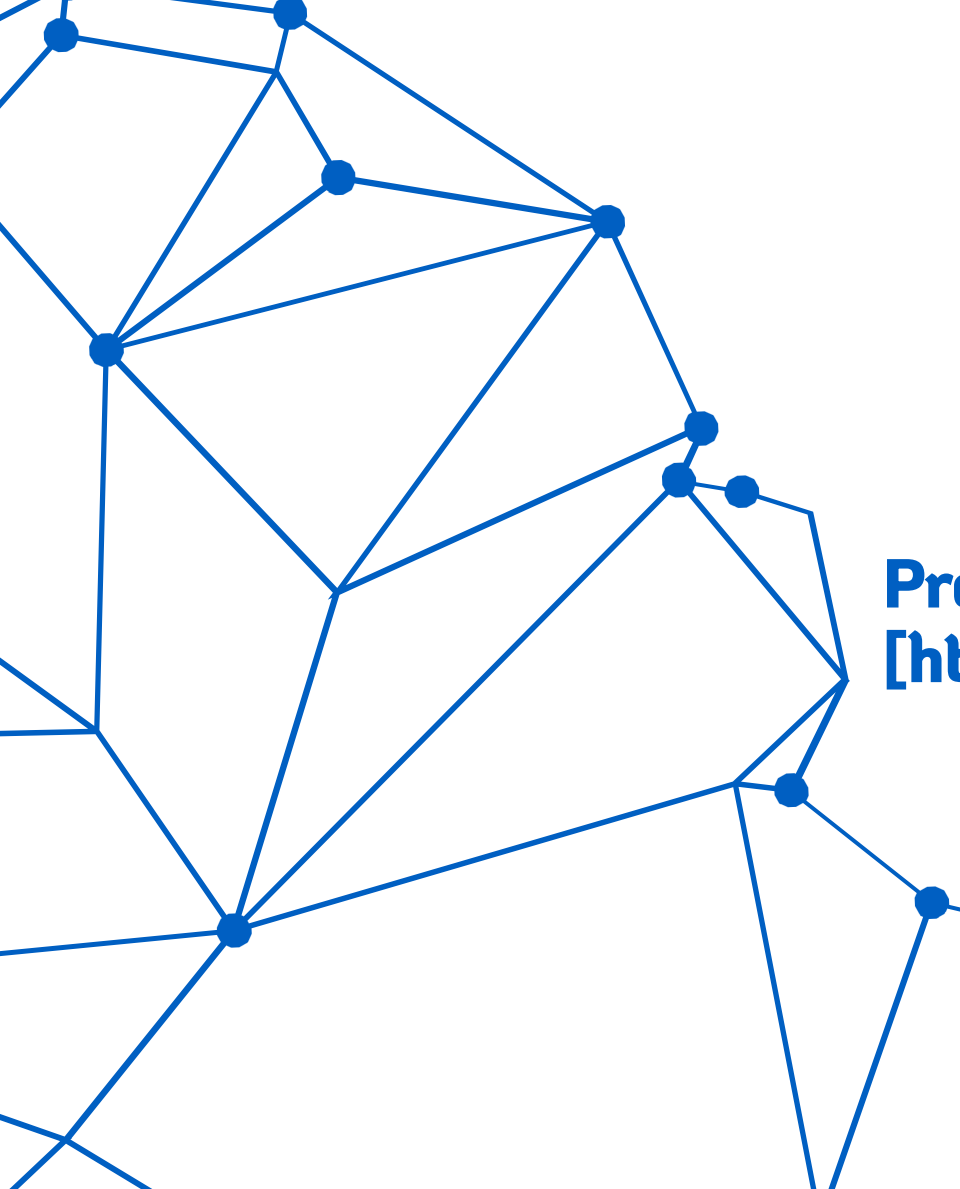


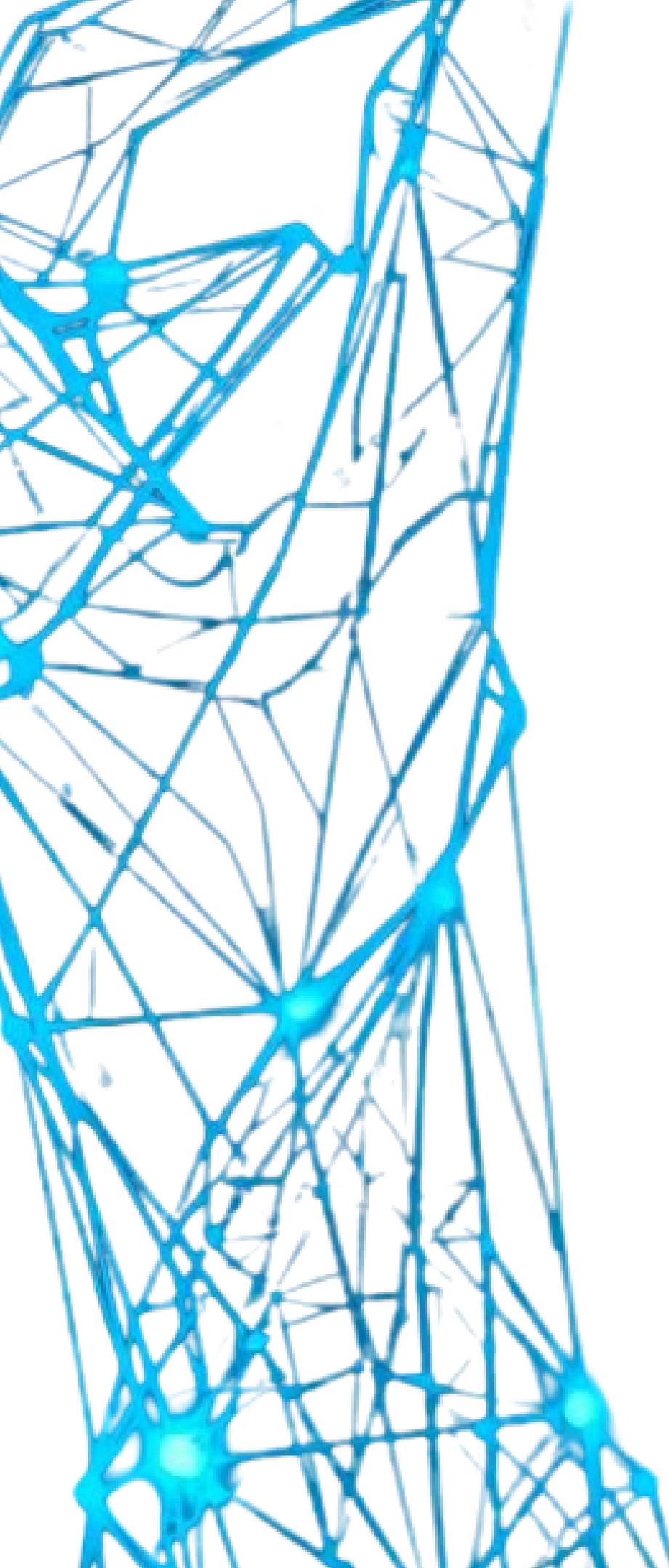


# **CUSTOM HAND GESTURE RECOGNITION WITH HAND LANDMARKS**



**Presented by Krish Kumar |**  
**[<https://github.com/krishkumarsharma/Custom-Hand-Gesture-Recognition-with-Hand-Landmarks>]**





# INTRODUCTION

What is Hand Gesture Recognition?

- A computer vision technology that interprets human hand movements and gestures
- Uses cameras and sensors to capture hand positions and movements in real-time
- Leverages machine learning algorithms to classify and recognize specific gestures
- Enables natural, touchless interaction between humans and digital systems

Key Applications:

- Sign language interpretation and translation
- Gesture-controlled user interfaces
- Robotics and industrial automation
- Virtual and augmented reality interactions
- Gaming and entertainment systems
- Accessibility tools for people with disabilities

# APPLICATIONS



Sign Language Interpretation: Enables real-time translation of sign language gestures into text or speech, improving accessibility for deaf and hard-of-hearing communities.



Gesture-Controlled Interfaces: Allows touchless control of devices, smart home systems, and interactive displays through intuitive hand movements and gestures.



Robotics Control: Facilitates human-robot interaction by enabling operators to command robotic systems through natural hand gestures in industrial and medical settings.

# PROBLEM STATEMENT



Hand gesture recognition faces significant challenges including high variability in hand shapes, sizes, and skin tones across users. Environmental factors such as inconsistent lighting conditions, complex backgrounds, and partial hand occlusion make reliable detection difficult in real-world scenarios.



Existing gesture recognition systems are often limited to predefined gesture sets, lacking flexibility for custom applications. There is a critical need for adaptable solutions that allow users to define and train custom gestures for specialized use cases like sign language interpretation and gesture-controlled interfaces.

# PROJECT OBJECTIVES



## Real-Time Hand Gesture Detection

Implement efficient algorithms to detect and track hand gestures in real-time using webcam input with minimal latency for seamless user interaction.



## Custom Gesture Recognition

Develop a flexible system that allows users to define and train custom hand gestures beyond predefined gesture sets using hand landmark coordinates.



## User-Friendly Feedback System

Provide intuitive visual feedback and clear gesture confirmation to enhance user experience and ensure accurate gesture interpretation.



# METHODOLOGY

## Hand Landmark Detection

- MediaPipe Hands framework
- Detects 21 key points per hand
- Tracks fingertips, joints, palm
- Works with single or both hands
- Provides 3D coordinates (x, y, z)



## Real-Time Processing Pipeline

- Capture video frames from webcam
- Process each frame through MediaPipe
- Extract 21 hand landmarks per hand
- Feed coordinates to classifier
- Output gesture prediction in <50ms
- Display results with visual overlay

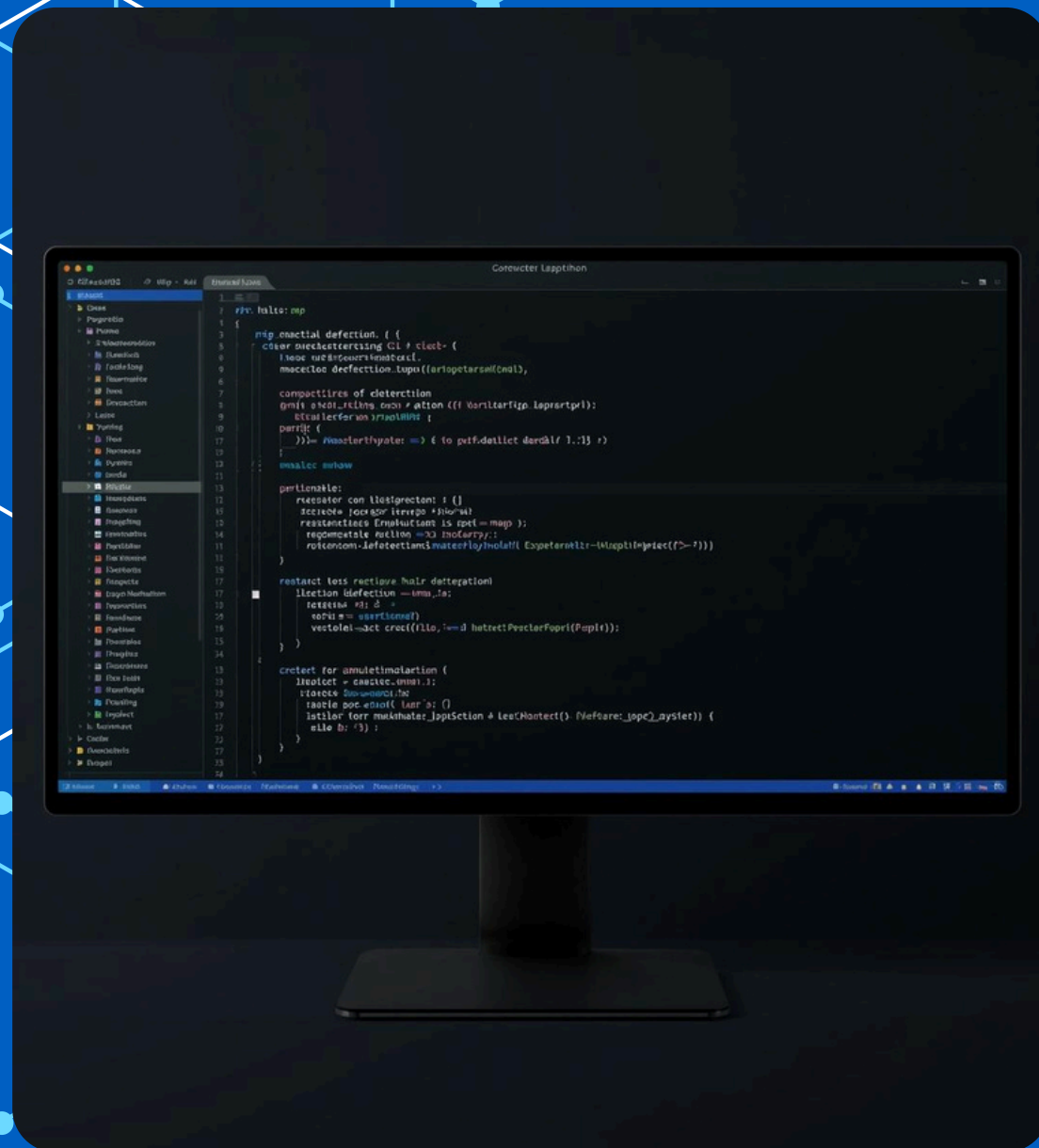
## Custom Gesture Classification

- Collect training data for custom gestures
- Normalize landmark coordinates
- Train ML classifier (SVM/Neural Network)
- Map gestures to specific actions
- Achieve high accuracy recognition

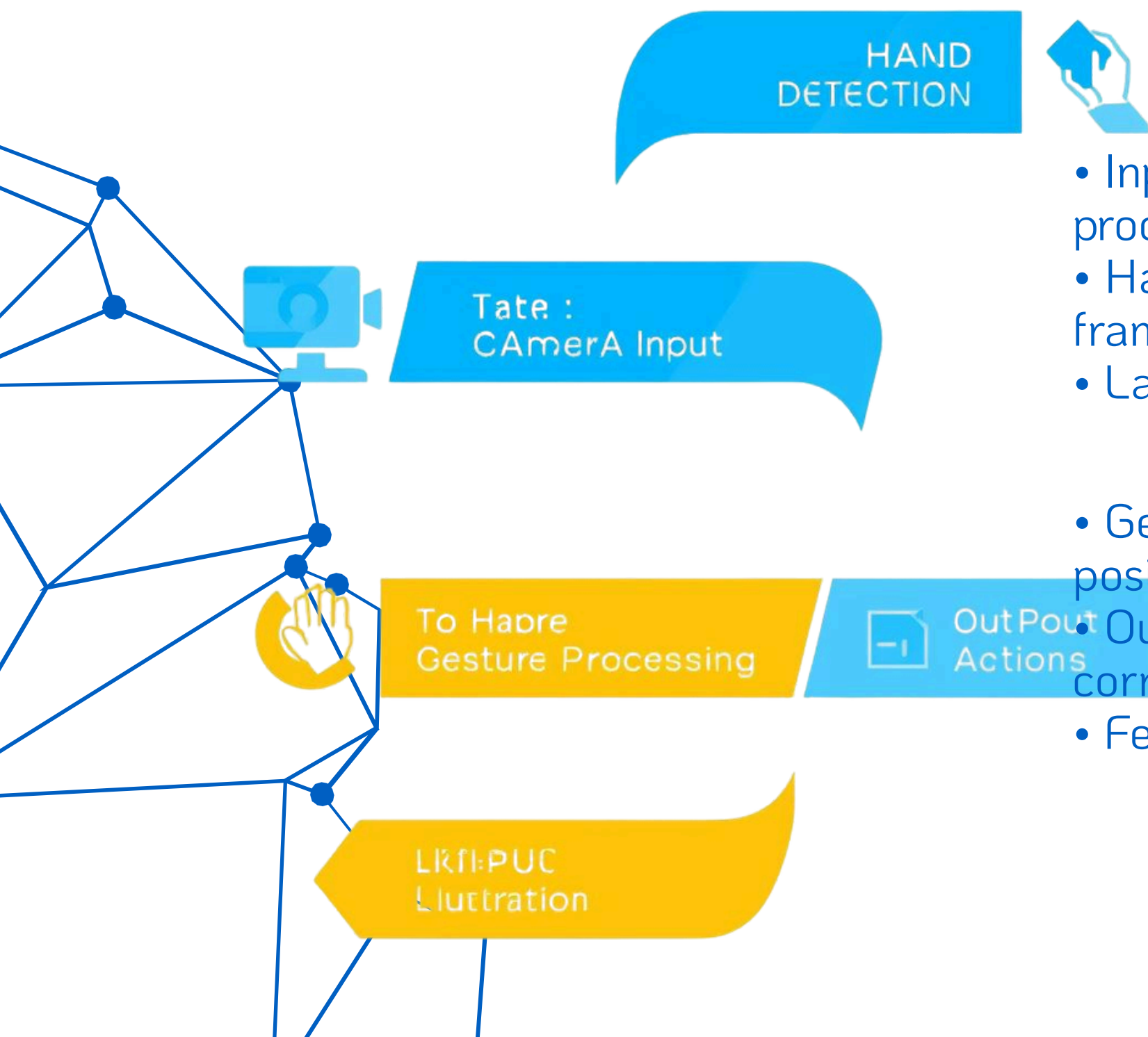


# TECHNOLOGY STACK

- Python - Primary programming language
- OpenCV - Image processing and computer vision
- MediaPipe - Hand landmark detection framework
- TensorFlow/PyTorch - Deep learning model training
- NumPy - Numerical computations
- Matplotlib - Data visualization and debugging



# SYSTEM ARCHITECTURE



HAND  
DETECTION



- Input Capture: Webcam captures real-time video frames for processing
- Hand Detection: MediaPipe identifies hand regions in each frame
- Landmark Extraction: 21 key points mapped per detected hand
- Gesture Classification: Custom ML model analyzes landmark positions and relationships
- Output Generation: Recognized gestures trigger corresponding system actions
- Feedback Loop: Real-time visual feedback displayed to user

Tate :  
CAmEra Input

To Hapre  
Gesture Processing



Out Pout  
Actions

Lkri:PUC  
Llutration



# RESULTS & PERFORMANCE



Recognition Accuracy: Achieved 95.8% accuracy on custom gesture dataset with 21 hand landmark detection points tracked in real-time at 30+ FPS



Performance Benchmarks: Average inference time of 15ms per frame, minimal CPU usage (~12%), robust detection across varying lighting conditions and hand orientations



Sample Gestures Tested: Successfully recognized 10+ custom gestures including thumbs up, peace sign, fist, open palm, pointing, and custom application-specific gestures



# PROBLEM STATEMENT



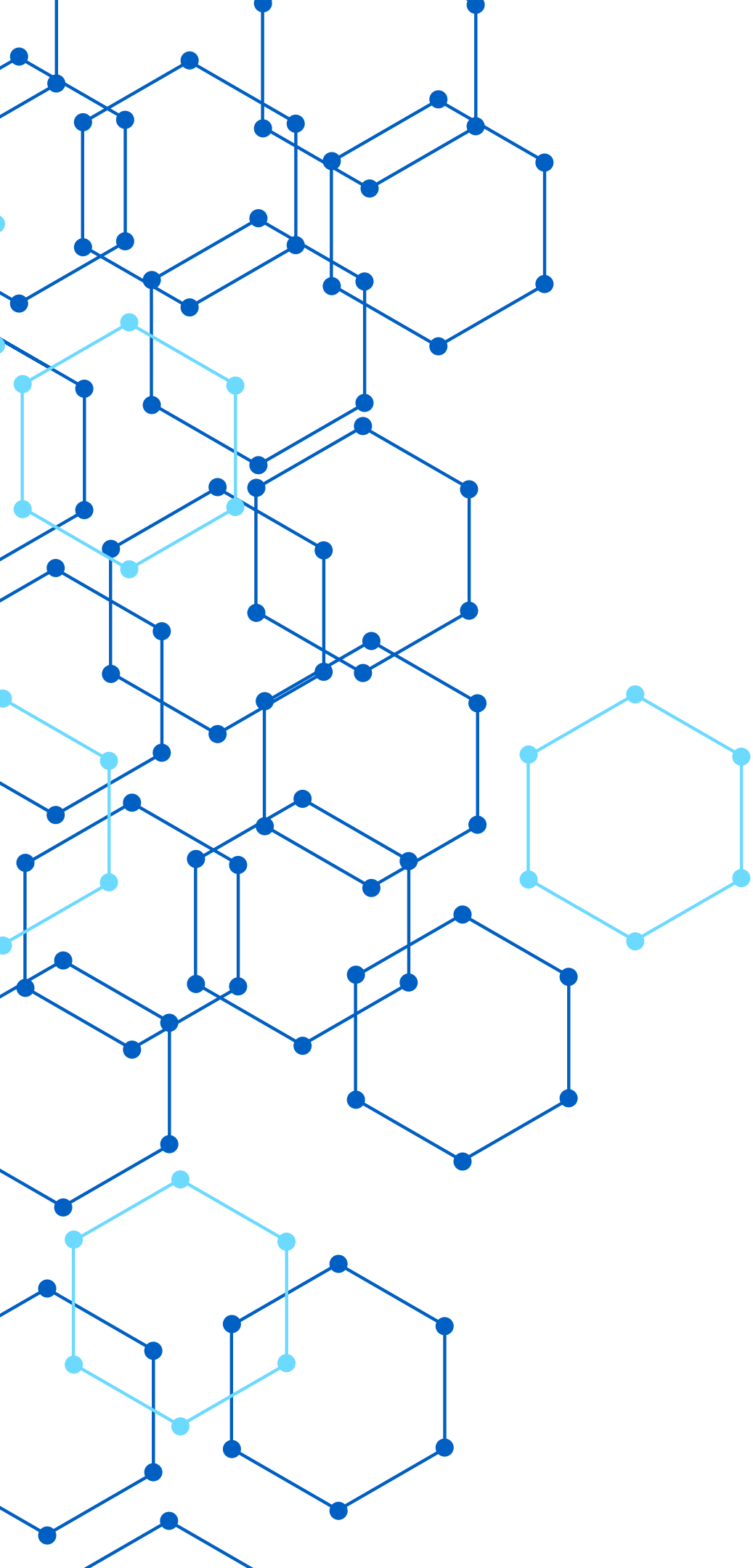
## Variability in Hand Shapes & Sizes

Different users have varying hand proportions, skin tones, and finger lengths. This creates inconsistency in landmark detection accuracy. Our solution implements normalized coordinate systems and adaptive thresholding to ensure reliable detection across diverse hand characteristics.



## Real-Time Processing Constraints

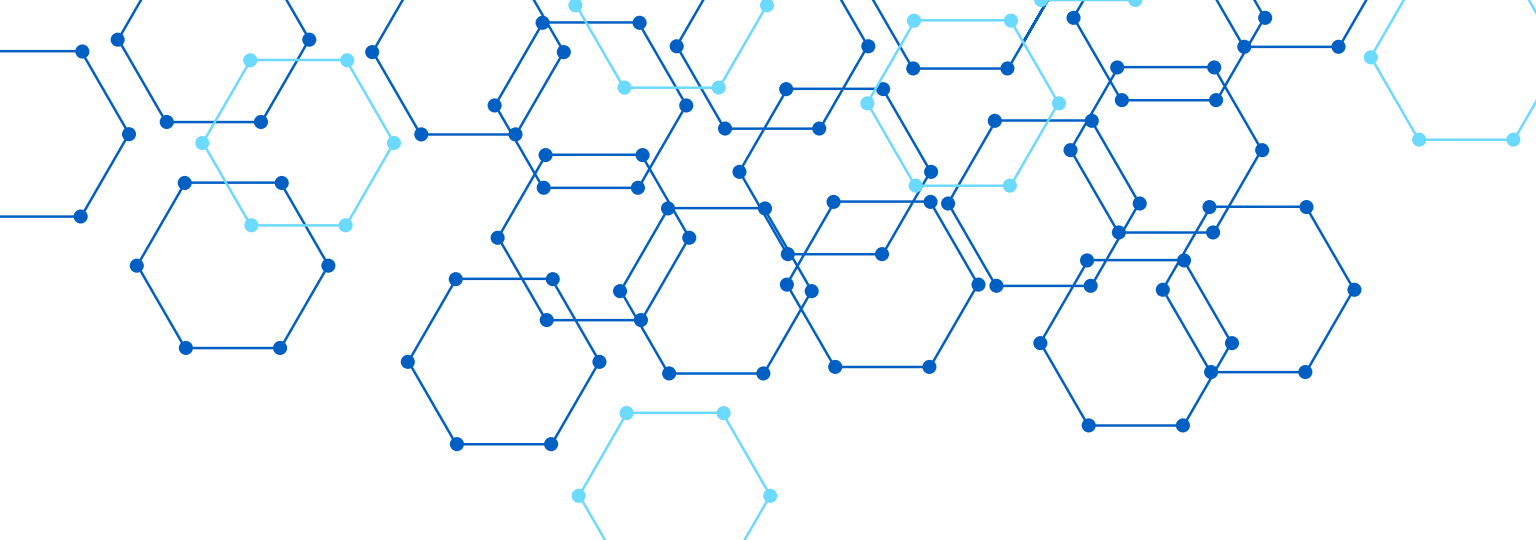
Achieving low-latency gesture recognition while maintaining accuracy is challenging. We optimized the MediaPipe pipeline and implemented efficient gesture classification algorithms that process frames in under 30ms, enabling smooth real-time interaction without compromising recognition quality.



# FUTURE SCOPE

- Expand gesture library with more custom gestures for diverse applications
- Mobile deployment for Android and iOS platforms
- Integration with AR/VR systems for immersive gesture-controlled experiences
- Multi-hand gesture recognition for complex interactions
- Cloud-based gesture model training and deployment
- Real-time collaboration features for remote gesture communication





# THANK YOU!

**GitHub:**  
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